Bronchial Asthma

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Definition & prevalence

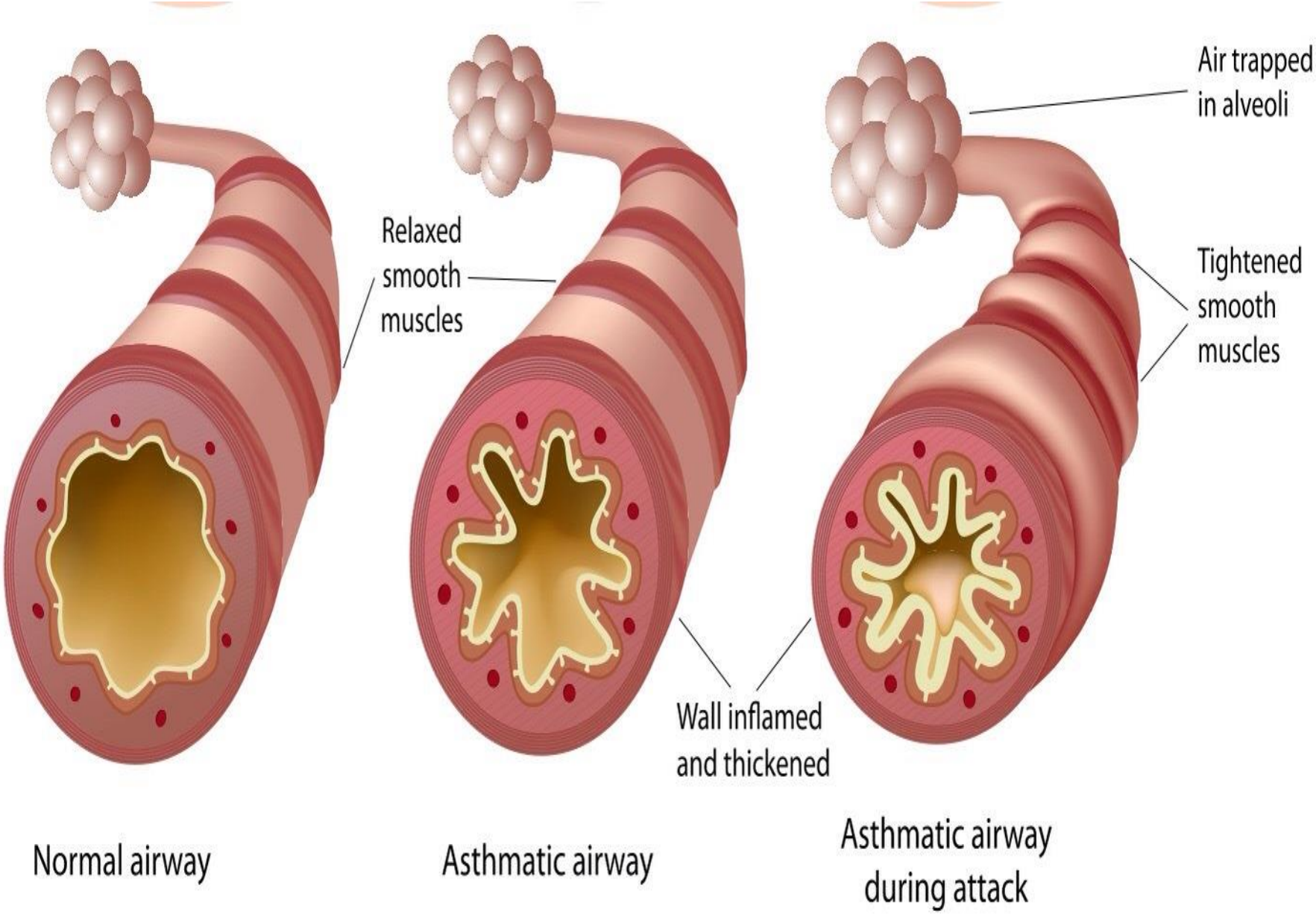
- Recurrent generalized reversible airway obstruction, manifested by paroxysmal dyspnea and wheeze.
- Affects 5—10% of the population

Aetiology & pathogenesis

- **Chronic inflammatory process** within the bronchial wall.
- **Cells**: eosinophils, neutrophils, lymphocytes, mast cells, macrophages.
- **Mediators**: histamine, prostaglandins, platelet activating factor, bradykinins, leukotrienes.
- **Bronchial hyper-responsiveness** (hyper-reactivity).

Aetiology & pathogenesis (cont.)

- **Bronchial narrowing** is caused by :-
  - Contraction of bronchial muscles (bronchospasm)
  - Swelling of mucous membrane
  - Plugging with viscid mucus
- **Diurnal variation**
- **Familial genetic predisposition**


Air trapped
in alveoli
Relaxed
smooth
muscles
Tightened
smooth
muscles
Wall inflamed
and thickened
Normal airway
Asthmatic airway
Asthmatic airway
during attack

Aetiology & pathogenesis (cont.)

- **Triggers of asthmatic attacks:-**

  - Allergens:-

  Mostly by inhalation as: pollens, house dust mite, fungal spores, feathers, animal dander.
  Ingested allergens are much less important.

  - Cold air
  - Smoke including tobacco smoke
  - Dust
  - Exacerbations of nasal allergy



Asthma Triggers

Aetiology & pathogenesis (cont.)

- **Triggers of asthmatic attacks (cont.):-**
  - Respiratory viral infection
  - Drugs: β-blockers, aspirin, NSAID
  - Physical exercise
  - Emotional stress
  - Hormonal : ↑ premenstrual, 1st trimester pregnancy
  - Occupational exposure : dust, organic material, fumes, chemicals.
  - Gastro-esophageal reflux.

COMMON CAUSES OF ASTHMA

some common causes and triggers that everyone should know

- POLLUTION
- SMOKING
- BACTERIA AND VIRUSES
- GENETIC
- PETS
- HOUSEHOLD CHEMICALS
- DUST

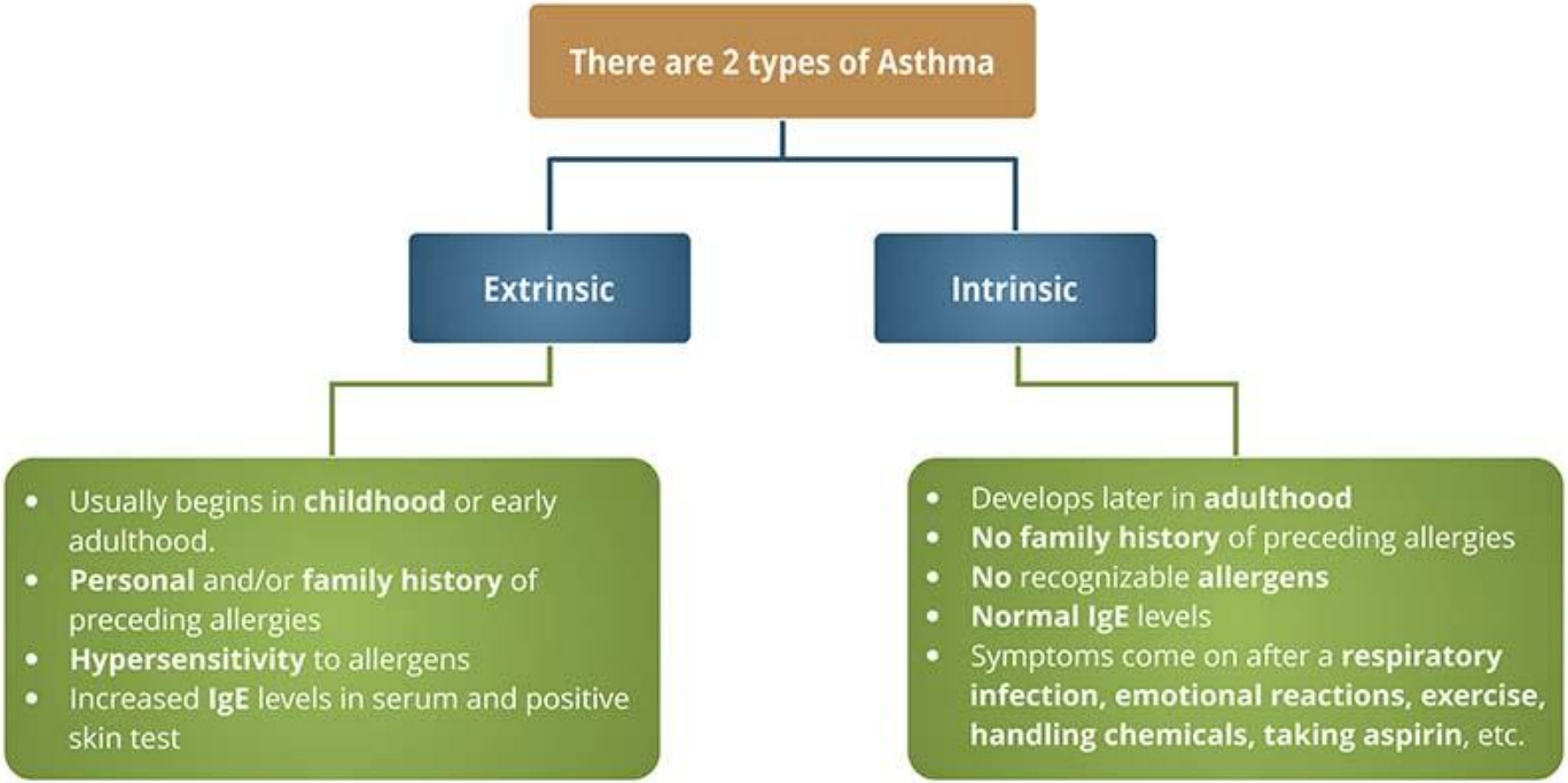


https://asthmabhawan.com/

Clinical picture: Clinical types

2 clinical types: changing classification over time:

Extrinsic	**Intrinsic**
Early onset	Late onset
Childhood onset	Adulthood onset
Atopic	Non-atopic
Episodic	Chronic
Intermittent	Persistent
Controlled	Uncontrolled, Partly controlled,


There are 2 types of Asthma
Extrinsic
Intrinsic
• Usually begins in childhood or early adulthood.
• Personal and/or family history of preceding allergies
• Hypersensitivity to allergens
• Increased IgE levels in serum and positive skin test
• Develops later in adulthood
• No family history of preceding allergies
• No recognizable allergens
• Normal IgE levels
• Symptoms come on after a respiratory infection, emotional reactions, exercise, handling chemicals, taking aspirin, etc.

Clinical picture: Symptoms

- **During the attack:**

➢ Starts sudden – acute – paroxysmal.
➢ More during night and early morning
➢ Dyspnea is the main symptom, more expiratory
➢ Wheeze: audible
➢ Cough:

- Dry during the attack
- Productive scanty tenacious sputum at the end.
- May be the only symptom specially in children and is called “Cough variant asthma“.

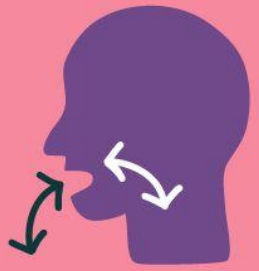

Coughing
Wheezing
Chest tightness
Shortness of breath

Clinical picture: **Symptoms (Cont.)**

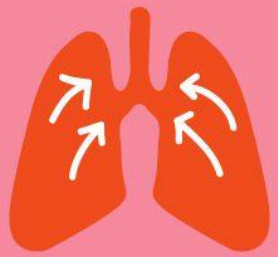
- The attack ends either:
  - Spontaneous
  - After use of bronchodilator
  - Persist: *"Severe acute asthma / Status Asthmaticus"*
  - Fatal

- **In between attacks:**
  - Intermittent asthma: free
  - Persistent asthma: persistent symptoms with diurnal variation

Severe Asthma Symptoms



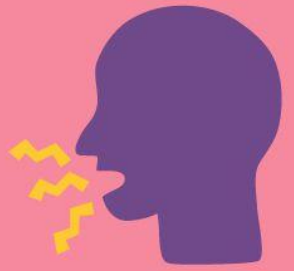
Rapid breathing



Shortness of breath



Chest tightness



Wheezing



Nighttime asthma episodes



Chronic cough



Racing heartbeat

verywell

Clinical picture:

Progress of asthma

- Childhood-onset asthma:-

  50 % grow out of asthma by age of 8 ys.

  25 % grow out of asthma by age of 12 ys.

  25 % persist into adulthood.

- Adulthood-onset asthma:

  Tends to be chronic

D.D. of bronchial asthma

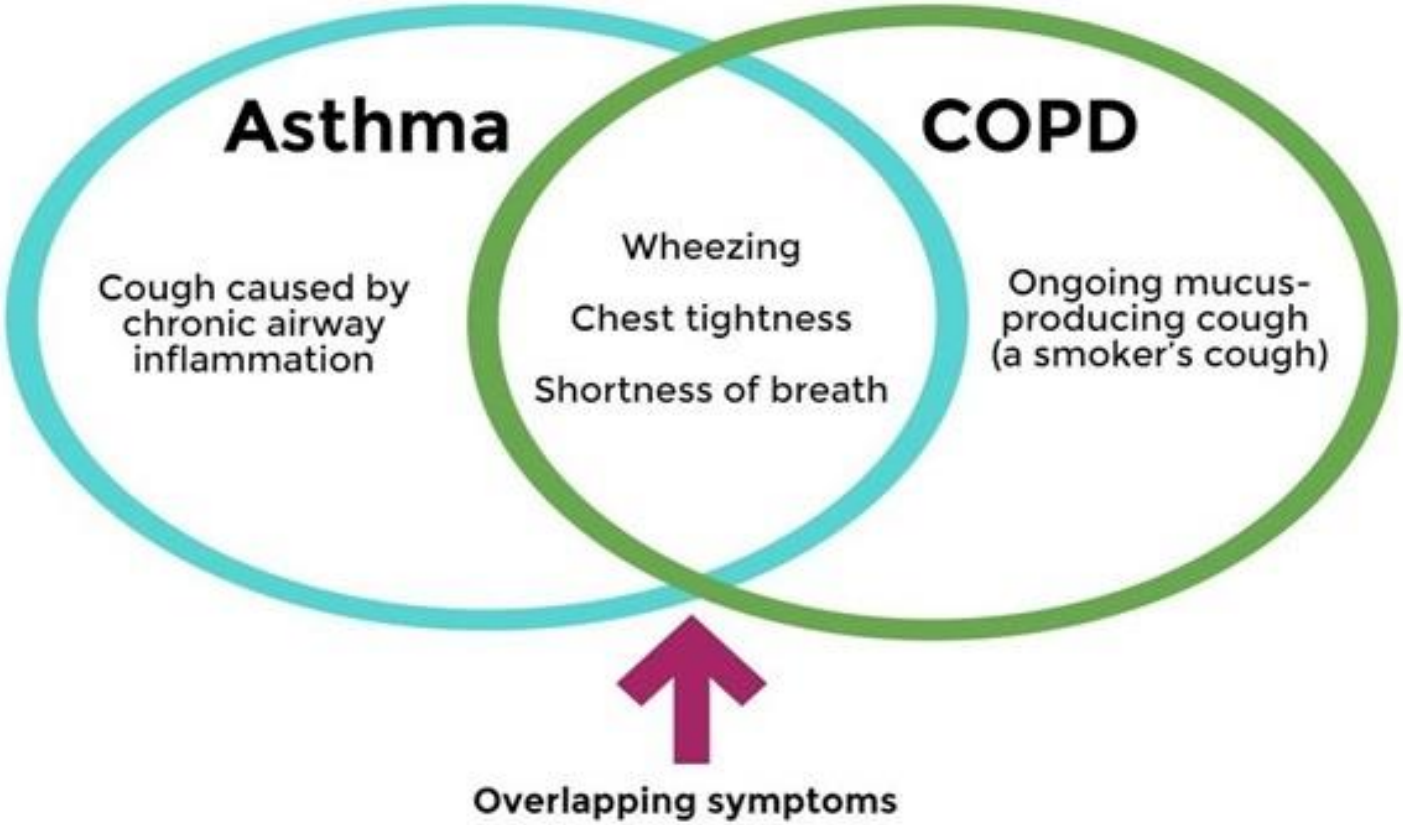
1- COPD (Ch. Bronchitis / emphysema):

- Age of onset: older in COPD
- Smoking: in COPD
- Main symptom:
  - Asthma:- paroxysmal dyspnea & wheeze
  - Ch. bronchitis:- cough and expectoration
  - Emphysema:- progressive exertional dyspnea
- Reversibility of airway obstruction: reversible in asthma

COPD Vs Asthma

	COPD	bronchial asthma
Smoker or ex-smoker	most	possibly
Symptoms under age 35	rare	common
Chronic productive cough	common	uncommon
breathlessness	Persistent and progressive	variable
Waking at night time with breathlessness or wheeze	uncommon	common
Significant diurnal or day to day variability of symptoms	Uncommon	common
FEV1 and FEV1/FVC ratio return to normal with drug therapy	Never with significant disease	probably

Asthma vs. COPD symptoms



D.D. of bronchial asthma (cont.)

2- Other causes of acute sudden dyspnea:

- Cardiac asthma (Lt heart failure).
- FB inhalation.
- Spontaneous pneumothorax.
- Pulmonary infarction and embolism.
- Hysterical, psychogenic.

Management of asthma

1. Education of the patient
2. Avoidance of triggering factors / desensitization
3. Available anti-asthma drugs
4. Management of an attack of asthma = quick relief medications
5. Management in between attacks = stepwise approach for long term management = preventive therapy = prophylactic therapy = anti-inflammatory ttt. = controller ttt.
6. Management of severe acute asthma

Management: 3. Available anti-asthma drugs

(A) Bronchodilators:

1- Sympathomimetics

2- Theophyllin

3- Anticholinergics

(B) Anti-inflammatory drugs:

1- Steroids

2- Antileukotrienes

3- Biological ttt

(A) **Bronchodilators**:

1- Sympathomimetics:

- Non selective: adrenaline
- Selective B2 agonists: act on β2 receptors of the bronchi.
- Available as:
  - Injection
  - Oral:
    Some are short acting (6h) as salbutamol and terbutaline.
    Some are long acting as Bambuterol
  - Inhalation: >>>>>>>>

Management Available anti-asthma drugs

(A) Bronchodilators: 1- Sympathomimetics (cont):

- Inhalation:

- Some are short acting (6h) as salbutamol and terbutaline.
- Some are long acting (12h) Long Acting Beta Agonists (**LABA**) as formoterol and salmeterol.
- Some are ultra-long acting beta agonists (24h) as Indacaterol and Vilanterol
- Inhalation therapy must be encouraged as it works rapidly within 5-10 minutes and avoids side affects of systemic use.

- Available as :-

  - Metered dose inhaler: but needs hand-breath coordination



  - Dry powder inhalers: but needs sufficient inspiratory flow







  - Nebulize



Management Available anti-asthma drugs

(A) Bronchodilators (cont.):

2- Theophyllin:-

- Act directly on smooth muscles of bronchi.
- Available as:
  - Injection
  - Oral: some are long acting sustained release
  - Suppository

Management Available anti-asthma drugs

(A) Bronchodilators (cont.):

3- Anticholinergics:

- Have atropine-like action
- Available as inhalation.
- Short acting Ipartropium bromide
- Long acting Tiotropium, Glycopyrronium, Umeclidinium (for COPD mainly)

(B) Anti-inflammatory drugs:

1- Steroids: Best anti-inflammatory drugs - Inhaled route is preferred - Avoid systemic use except in exacerbations.

Available as:-

- Injection:

- short acting (hours): Hydrocortisone – Dexamethasone.
- long acting (days/weeks): Methylprednisolone acetate, Triamcinolone, Betamethasone dipropionate

- Oral: Prednisolone
- Inhalation (inhaled cortico-steroids) (**ICS**):

- effective and avoids side effects of systemic steroids

- As: Beclomethasone, Budesonide, Fluticasone, ciclesonide

Management Available anti-asthma drugs

2- Antileukotrienes:

As Montelucast. Oral, so considered an alternative in patients who refuse or are unable to use an inhaler. Used starting from age 6 months. Alternative or add-on to ICS.

3- Biological treatment:

As the monoclonal antibodies: Omaliuzumab anti IgE, and Dupilumab anti interleukin-4 and 13 (IL-4 & IL-13) cytokines. May help in selected subgroup of "difficult to treat asthma" (personalized medicine). Given only if still uncontrolled using all bronchodilators and other anti-inflammatory drugs. Very expensive.

Management 4. Management of an attack

Management of an attack of asthma,

Quick relief medications,

Reliever treatment.

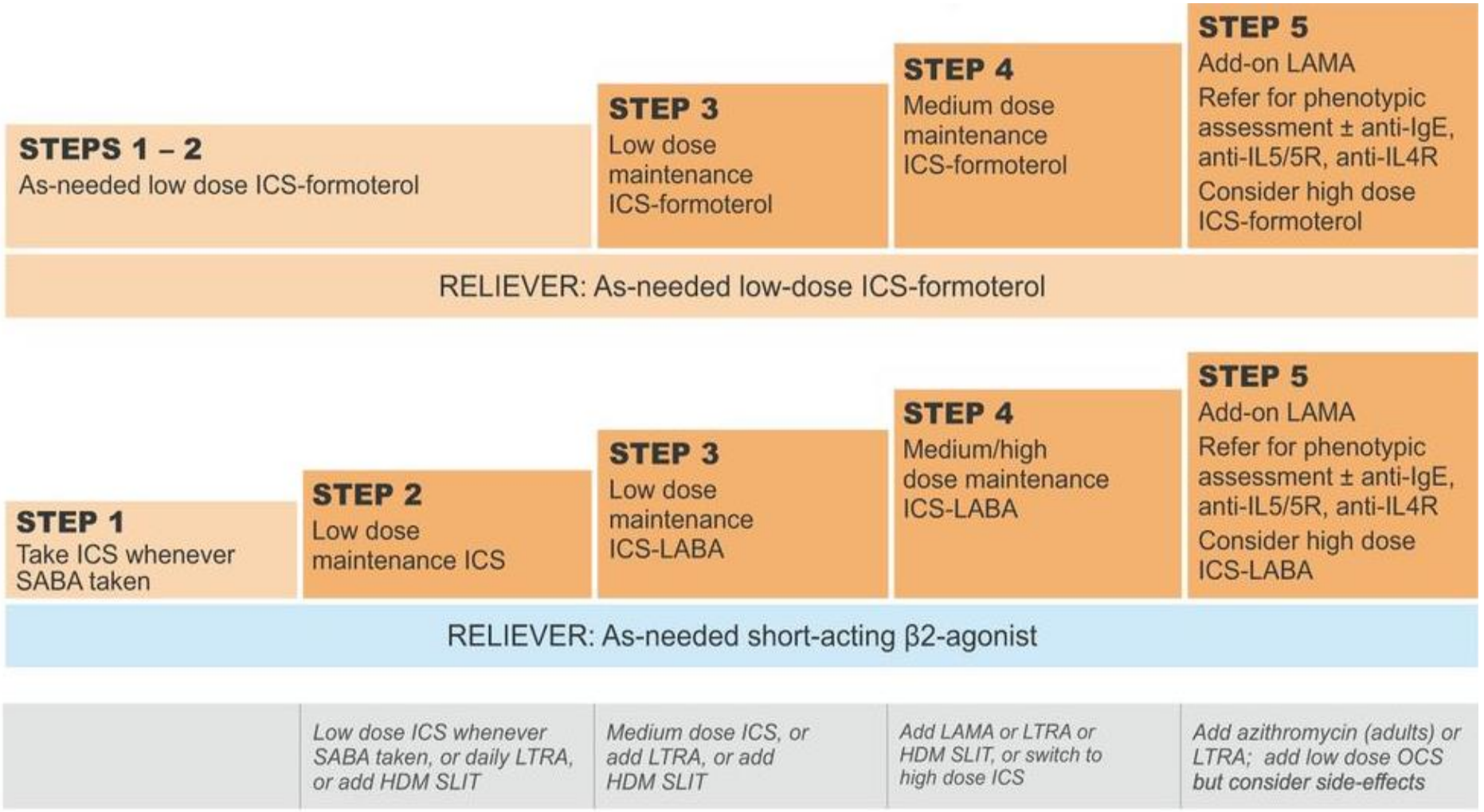
Use inhaled rapidly / short acting B2 agonists

Management 5. Treatment in between attacks

- **Treatment in between attacks,**
- **Stepwise approach for long term management,**
- **Preventive therapy of asthma,**
- **Prophylactic treatment,**
- **Anti-inflammatory treatment,**
- **Controller treatment.**

1. Mainly use anti-inflammatory drugs mentioned before.
2. In addition, if necessary, to long acting bronchodilators.
3. Do not use long acting bronchodilators alone without ICS.

GINA 2023 for adults

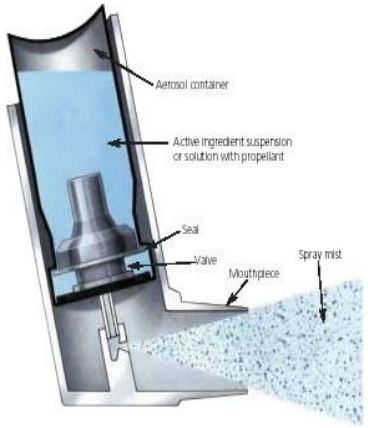


Management 6. Management of severe acute asthma

1. Hospitalization
2. Reassurance
3. Oxygen therapy: humidified, high concentration in pure asthma
4. Selective β_2 agonist by nebulizer or MDI connected to a spacer
5. Aminophyllin IV 250 mg slowly.
6. Ipratropium bromide by nebulizer or MDI connected to a spacer.

7. Systemic steroids: immediate IV hydrocortisone 200 mg and/or 30-60 mg oral prednisolone.
8. Magnesium sulphate infusion
9. Hydration.
10. Electrolyte correction.
11. Antibiotics only if there is evident bacterial infection.
12. ICU admission and mechanical ventilation in special severe situations.

How to use metered dose inhaler MDI



1) Shake well (4-5 times) then take the cap off.
2) Prime the inhaler (if first time or left for > 1week)
3) Exhale slowly, as far as comfortable not forcibly.
4) Make head slightly extended
5) Hold inhaler upright between fingers above and thumb on base.
6) Immediately place the inhaler in the mouth between the teeth, with the tongue flat under the mouthpiece. Form a good seal with the lips.
7) Start to inhale slowly, through the mouth and at the same time press the canister to actuate a dose (hand-breath coordination).
8) Maintain slow and deep inhalation, through the mouth, until lungs are full of air. This takes an adult 4–5 s (child 2-3 s).
9) At the end inhalation, take inhaler out and close the lips.
10) Continue hold breath for as long as 10 seconds, then exhale slowly.
11) If another dose is required wait at least 30 sec.

How to use dry powder inhaler DPI







1) Follow the dose preparation instructions in the PIL.

2) Do not point the mouthpiece downwards once a dose has been prepared for inhalation because the dose could fall out.

3) Exhale slowly, as far as comfortable. Do not exhale into the DPI

4) Start to inhale forcibly through the mouth from the very beginning. Do not gradually build up the speed of inhalation.

5) Continue inhaling until the lungs are full.

6) At the end of the inhalation take the inhaler out of the mouth

7) Continue holding breath for 10 sec. then exhale slowly


Thank
you